CERAMIC BALLS - PROPERTIES OF MATERIALS

PHYSICAL PROPERTIES	SAPPHIRE AND RUBY	ALUMINA OXIDE	SILICON NITRIDE	ZIRCONIUM OXIDE
STRUCTURE	single crystal	multi-crystal	multi-crystal	multi-crystal
CHEMICAL FORMULA	$\grave{A}l_2\hat{I}_3$	$\grave{A}l_2\hat{I}_3$	Si_3N_4	ZrO_2
PURITY %	99.99	99.8	95.00	97.00
DENSITY g/cm$_3$	3.99	3.90	3.20	5.50
OPERATING TEMPERATURE °C		1800	1100	1000
MELTING POINT	2050 °C	2050 °C	1900 °Ñ	
SOFTENING POINT	1800 °C	1725 °Ñ	1400 °Ñ	
SPECIFIC HEAT AT 25 °C (CAL/g/°C) THERMAL CONDUCTIVITY	0.18 36 W/m°k	0.25 29 W/m°k	0.17 29 W/m°k	9 W/m°k
MECHANICAL PROPERTIES	SAPPHIRE AND RUBY	ALUMINA OXIDE	SILICON NITRIDE	ZIRCONIUM OXIDE
VICKERS Hv10 HARDNESS (N/mm2)	17000	16500	24000	20000
MODULUS OF ELASTICITY (N/mm2)	$4,3\tilde{o}10^5$	$3,5\tilde{o}10^5$	$3,1\tilde{o}10^5$	$2\tilde{o}10^5$
BREAKING MODULUS AT 25 °C (N/mm2)	392	470	700	600
COMPRESSIVE STRENGTH AT 25 °C (N/mm2)	2060	2354	2500	2100

CHEMICAL RESISTANCE

SAPPHIRE / RUBY:
inert to most acids at very high temperatures.

ALUMINA (OXIDE):
inert to most acids, but not recommended in environments with hydrochloric or hydrofluoric acids or strong alkaline solutions

SILICON NITRIDE:
inert to most acids.

ZIRCONIUM OXIDE:
inert except to hydrofluoric acid and strong concentrations of sulphuric acid.